General Biology 1

First Semester

Supplementary Materials

Module 5

Phases of Cell Cycle and their Control Points

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What I Know

Directions: Read each situation carefully and choose the best answer.

1. Which phase of the cell cycle involves DNA replication?
   a. G1 phase b. S phase
   c. G2 phase d. M phase
2. What happens during the G1 phase?
   a. DNA is copied
   b. Chromosomes separate
   c. The cell grows and performs normal functions
   d. The nucleus divides
3. What is the main purpose of cell cycle checkpoints?
   a. To speed up cell division b. To produce energy for the cell
   c. To ensure proper cell division d. To store genetic material
4. The G1 checkpoint primarily checks for:
   a. Proper chromosome separation b. DNA damage and cell size
   c. Completion of mitosis d. Spindle fiber attachment
5. During which phase do chromosomes align at the center of the cell?
   a. Prophase b. Metaphase
   c. Anaphase d. Telophase

What I Need to Know

This supplementary module is designed to help you understand the **phases of cell cycle and their control points**. In this lesson, you will explore how the importance of cell cycle control points and identify key events in each phase.

Lesson 5 Phase of Cell Cycle and Their Control Points

The cell cycle is a highly organized series of events that cells go through as they grow, replicate their DNA, and divide into two new cells. To ensure accurate cell division, the cell cycle is regulated by specific control points (checkpoints). These control mechanisms help prevent errors, maintain genetic stability, and protect the organism from conditions such as uncontrolled cell growth.

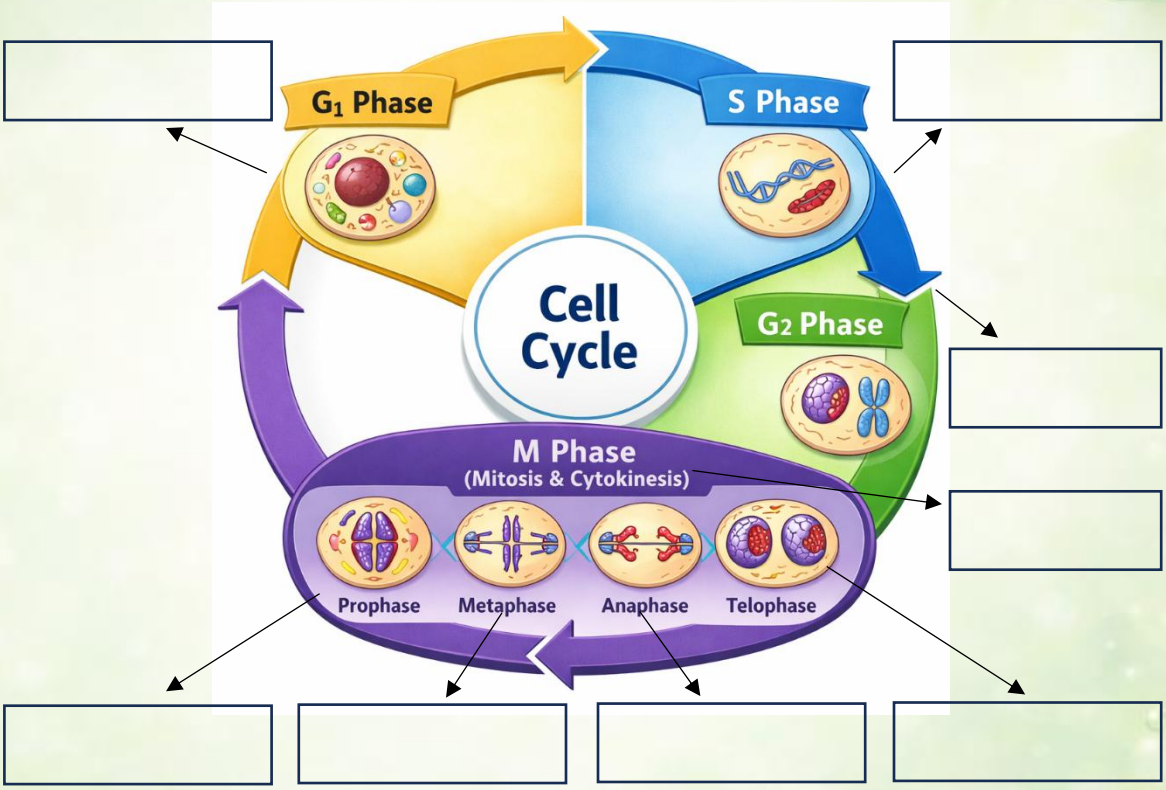
After going through this supplementary module, you are expected to:

1. describe the phases of the cell cycle (interphase, M-phase, and cytokinesis) and identify their key events;
2. sequence the stages of the cell cycle and analyze the role of each checkpoint (G1, G2, and M) in regulating cell division; and
3. explain the importance of cell cycle control points by predicting the consequences of checkpoint failure on cell function.

What's In

Activity 1: "Phase Finder: The Cell Cycle Challenge"

Directions: Match the event happened in cell cycle to its corresponding phase. Use the events inside the box.



- DNA is copied to produce identical sister chromatids.
- Cell grows, produces organelles, and carries out normal functions.
- Chromosomes align along the center of the cell.
- Cytoplasm divides, forming two daughter cells.
- Nuclear membranes reform around the separated chromosome sets.
- Spindle fibers attach to chromosomes and move them toward poles.
- Chromosomes condense and become visible under a microscope.
- Cell prepares for mitosis by producing proteins and checking DNA for errors.

The cell cycle is a continuous process by which a cell grows, replicates its DNA, and divides to form two identical daughter cells. It ensures proper growth, repair, and reproduction in living organisms. The cycle is divided into two major parts: Interphase (G_1, S, G_2) and the M Phase (mitosis and cytokinesis).

The **interphase** of the cell cycle consists of the G_1, S, and G_2 phases, during which the cell grows, replicates its DNA, and prepares for division. In the **G_1 phase**, the cell increases in size, produces organelles such as mitochondria and ribosomes, and carries out normal metabolic functions like protein synthesis. It also accumulates energy and synthesizes RNA and proteins needed for DNA replication. During this stage, the cell checks whether conditions are favorable for division and may enter a resting stage (G_0) if conditions are not suitable.

In the **S phase**, DNA replication occurs, where the genetic material is copied to produce identical sister chromatids. Each chromosome duplicates, forming two identical strands joined at the centromere. Histone proteins are produced to help package the newly replicated DNA, and centrosomes begin to duplicate. Although the DNA content doubles, the number of chromosomes remains the same. The cell also uses proofreading mechanisms to ensure that errors during DNA replication are minimized.

In the **G_2 phase**, the cell continues to grow and prepares for mitosis. It synthesizes proteins necessary for cell division, including those involved in forming spindle fibers. Organelles and other cellular components are further produced, and microtubules begin to form. Importantly, the cell checks the replicated DNA for any errors and repairs them to ensure accurate division. This phase serves as the final checkpoint before the cell enters mitosis, ensuring that all processes from earlier stages have been completed correctly.

During the **M phase** of the cell cycle, the cell undergoes mitosis followed by cytokinesis, resulting in two identical daughter cells. In **prophase**, the chromosomes condense and become visible under a microscope, with each chromosome consisting of two sister chromatids joined at the centromere. The nuclear membrane begins to break down, the nucleolus disappears, and centrosomes move toward opposite poles of the cell. At the same time, spindle fibers start to form and begin attaching to the chromosomes.

In **metaphase**, the chromosomes align along the center of the cell, known as the metaphase plate. Spindle fibers are fully attached to the centromeres (kinetochores) of each chromosome, ensuring that each sister

chromatid is connected to opposite poles. The cell checks that all chromosomes are properly aligned before proceeding to the next stage.

During **anaphase**, the sister chromatids separate at the centromere and are pulled toward opposite poles of the cell by the spindle fibers. As this occurs, the cell elongates, and each chromatid is now considered an individual chromosome. In **telophase**, the chromosomes reach the opposite poles and begin to decondense back into chromatin. Nuclear membranes reform around each set of chromosomes, nucleoli reappear, and the spindle fibers start to break down.

Finally, in **cytokinesis**, the cytoplasm divides to form two separate daughter cells. In animal cells, a cleavage furrow forms and pinches the cell membrane, while in plant cells, a cell plate forms and develops into a new cell wall. Each daughter cell contains identical genetic material and enters the G_1 phase, completing the cell cycle.

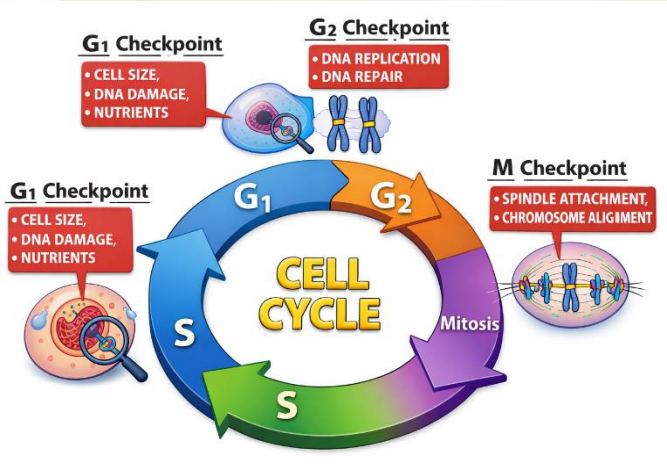
What's New

Activity 2: "Checkpoint Analysis"

Directions: Analyze the given situation and determine which checkpoint it happens.

SCENARIO	WHICH CHECKPOINT FAILED?	ANALYSIS
A cell with damaged DNA continues to the S phase.		What should have happened? _____ _____ _____
DNA replication is incomplete, but the cell enters mitosis.		Why is this dangerous? _____ _____ _____
Chromosomes are not aligned properly, but the cell proceeds to anaphase.		What could happen to the daughter cells? _____ _____ _____
A cell lacks sufficient nutrients but continues to DNA replication.		What should the cell have done? _____ _____ _____

DNA replication errors were not corrected before mitosis.		Why is this harmful? _____ _____ _____
Spindle fibers did not attach properly to some chromosomes, but the cell proceeded.		What might happen to the chromosomes? _____ _____ _____
The cell grows too small and divides immediately without proper preparation.		Why is cell size important? _____ _____ _____
Chromosomes fail to separate equally during anaphase.		What condition may result? _____ _____ _____
The cell completes DNA replication but skips checking for errors.		Possible outcome: _____ _____ _____
The cell proceeds to division even though spindle fibers are missing.		What will likely happen to the daughter cells? _____ _____ _____



The cell cycle checkpoints are control mechanisms that ensure a cell only proceeds to the next stage when conditions are correct and safe. These checkpoints act like "quality control points" during the cell cycle, specifically at the **G_1 checkpoint, G_2 checkpoint, and M (metaphase) checkpoint.** Each checkpoint monitors key events in the cell and prevents errors that could lead to damaged or abnormal cells.

At the **G_1 checkpoint**, the cell evaluates whether it is ready to begin DNA replication. During this stage, the cell must have enough nutrients, sufficient size, and undamaged DNA. This is also when the cell is actively growing, producing organelles, and carrying out normal functions. If DNA damage is detected or conditions are unfavorable, the checkpoint stops the cycle to allow repair or may direct the cell to a resting stage. If this checkpoint fails, damaged DNA may be copied during the S phase, which can lead to mutations or diseases such as cancer.

The **G_2 checkpoint** occurs after DNA replication in the S phase. At this point, the cell ensures that DNA has been completely and accurately copied into identical sister chromatids. It also checks for any replication errors and prepares for mitosis by producing proteins and necessary structures like spindle fibers. If DNA is incomplete or damaged, the checkpoint halts progression to allow repairs. Failure of the G_2 checkpoint may result in cells entering mitosis with errors, leading to improper chromosome separation and defective daughter cells.

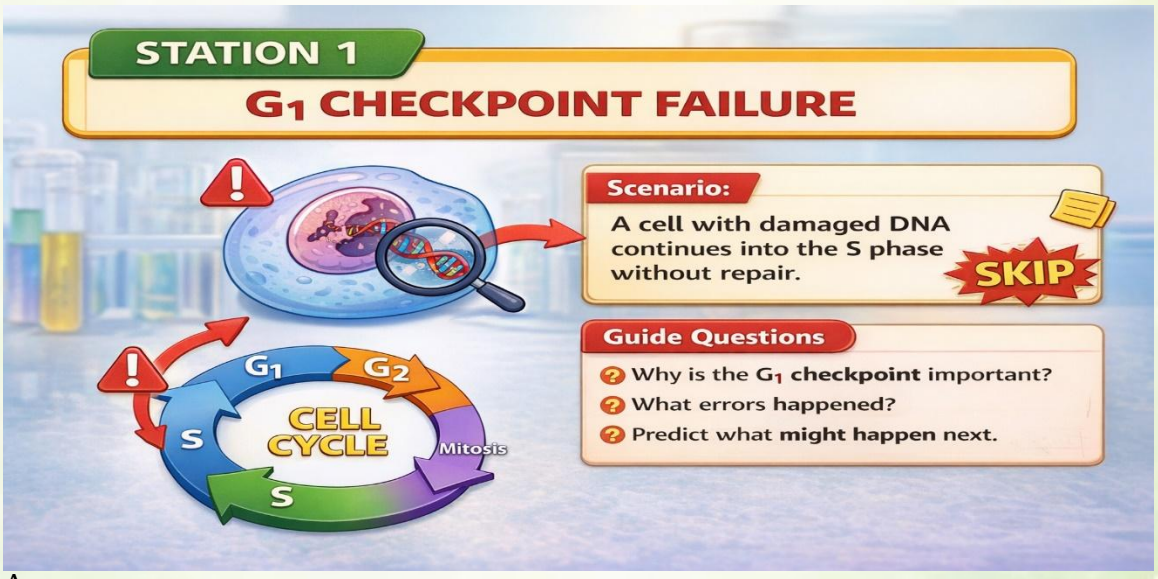
The **M checkpoint**, also known as the spindle or metaphase checkpoint, takes place during metaphase of mitosis. Here, the cell verifies that all chromosomes are properly aligned at the center (metaphase plate) and that spindle fibers are correctly attached to each chromosome. These spindle fibers are responsible for pulling sister chromatids apart during anaphase. If chromosomes are not properly attached or aligned, the checkpoint prevents the cell from proceeding. If this checkpoint fails, chromosomes may not separate equally, resulting in daughter cells with abnormal chromosome numbers (a condition known as aneuploidy).

Overall, these checkpoints are essential for maintaining the accuracy of cell division. They ensure that DNA is correctly replicated, chromosomes are properly distributed, and each daughter cell receives the correct genetic information. When checkpoints fail, it can lead to serious consequences such as genetic mutations, uncontrolled cell division, or cell death.

What's More

Activity 3: "Checkpoint Breakdown: Predict the Consequences"

Directions: This activity consists of three (3) stations your task is to answer the guide questions each station pertaining to the error occurred in each checkpoint.

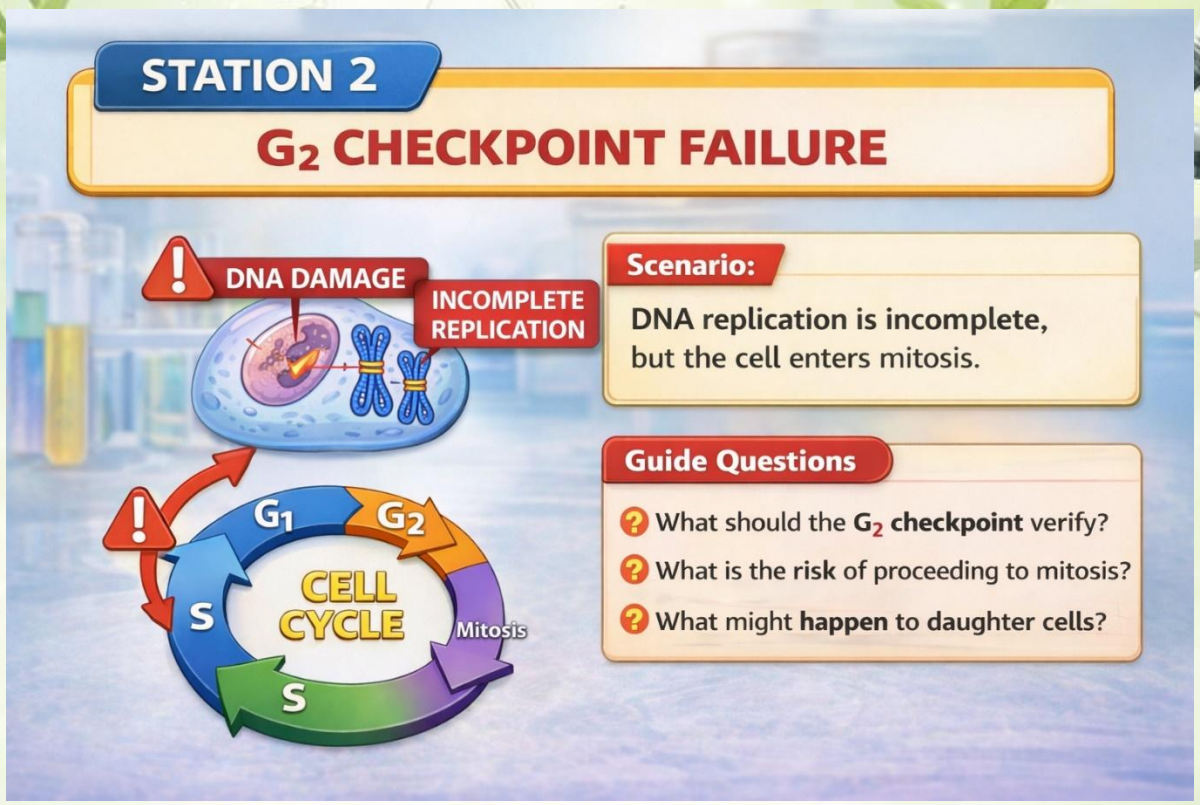


Answer:

1. _____
_____
_____

2. _____
_____
_____

3. _____
_____
_____

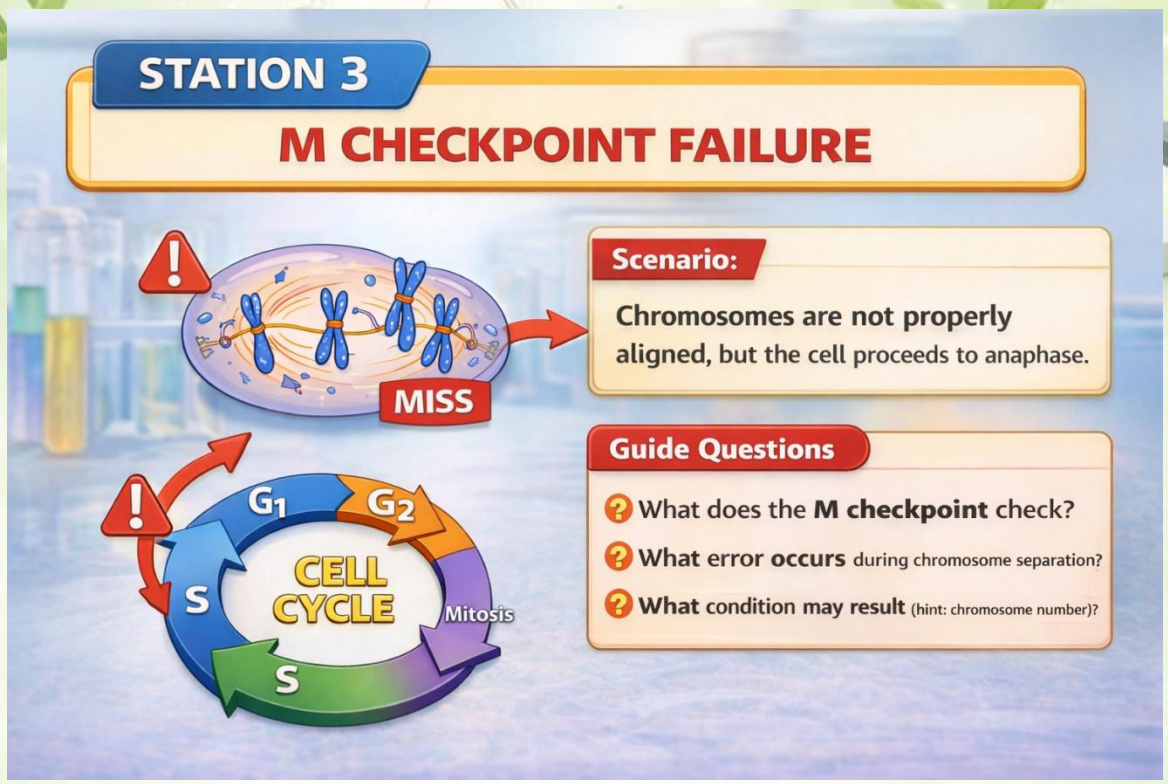

STATION 2
G2 CHECKPOINT FAILURE
DNA DAMAGE
INCOMPLETE REPLICATION
G1
G2
S
CELL CYCLE
Mitosis
S
Scenario:
DNA replication is incomplete, but the cell enters mitosis.
Guide Questions
What should the G2 checkpoint verify?
What is the risk of proceeding to mitosis?
What might happen to daughter cells?

Answer:

1. _____
_____
_____

2. _____
_____

3. _____
_____



Answer:

1. _____
_____
_____

2. _____
_____

3. _____
_____

To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A researcher observes that a cell with damaged DNA continues to replicate and divide uncontrollably. Which checkpoint most likely failed, and why?

a. M checkpoint, because chromosomes were not aligned
b. G_1 checkpoint, because DNA damage was not detected before replication
c. G_2 checkpoint, because spindle fibers did not form
d. Cytokinesis checkpoint, because cytoplasm did not divide

2. If a cell enters mitosis with incomplete DNA replication, what is the most likely consequence?

a. Faster cell division
b. Equal distribution of chromosomes
c. Formation of genetically identical daughter cells
d. Production of abnormal daughter cells with missing or damaged DNA

3. During an experiment, spindle fibers fail to attach to chromosomes, yet the cell proceeds to anaphase. What will most likely occur?

a. Chromosomes will duplicate again
b. Chromosomes will not separate properly
c. DNA replication will restart
d. Cell growth will stop permanently

4. A mutation disables the protein responsible for stopping the cell cycle when errors are detected. What is the most likely long-term effect?

a. Cells will stop dividing completely
b. Cells will divide only during G_1 phase
c. Cells will divide uncontrollably, possibly leading to cancer
d. Cells will skip cytokinesis only

5. Why is the M checkpoint considered critical for maintaining chromosome number stability?

a. It ensures DNA replication is complete
b. It allows the cell to grow properly
c. It checks chromosome alignment before separation
d. It produces proteins needed for mitosis